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Low Bioavailability and High TMAO Production: Novel Insights Into Acetylcarnitine and Carnitine Metabolism

Kristaps Krims-Davis[1,2] | Melita Ozola[1] | Valerija Razzivina[1,2] | Baiba Gukalova[1,2] | Ilze Konrade[2,3] | Maija Dambrova[1,2] | Edgars Liepinsh[1,2]

[1]Latvian Institute of Organic Synthesis, Riga, Latvia | [2]Riga Stradiņš University, Riga, Latvia | [3]Riga East Clinical Hospital, Riga, Latvia

Correspondence: Kristaps Krims-Davis (kristaps.krims-davis@farm.osi.lv)

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ABSTRACT

The food supplement acetylcarnitine is marketed for use in neurological support; however, research on its bioavailability and metabolic fate has been limited. This study investigated the absorption, metabolism, and excretion of acetylcarnitine compared with those of carnitine.

Healthy volunteers received either carnitine or an acetylcarnitine supplement (0.5 or 1.5 g). Plasma and urine samples were collected at baseline and at multiple time points (1–48 h) post intake and analyzed using LC–MS/MS. Both carnitine and acetylcarnitine exhibited low intestinal absorption and renal reabsorption. The peak plasma concentrations increased over the baseline values by 48% (acetylcarnitine) and 43% (carnitine) following a 1.5 g dose. However, the increase in area under the curve (ΔAUC) from acetylcarnitine was 7.7-fold lower than that from carnitine. Elevated plasma levels of carnitine and acetylcarnitine led to a 5-fold increase in clearance, and a substantial portion of the supplements were excreted via urine. The acetylcarnitine supplement was mostly eliminated in the form of carnitine. Approximately 90% of both supplements were metabolized to TMAO, reaching 50 μM in plasma—levels previously found to be associated with adverse health outcomes.

Acetylcarnitine has significantly lower bioavailability than carnitine. The intake of both supplements resulted in substantial TMAO production, raising potential health concerns.

1 | Introduction

Carnitine is an essential nutrient for transporting long-chain fatty acids across the mitochondrial membrane, enabling their beta-oxidation and the subsequent production of adenosine triphosphate (ATP), the primary energy source in the human body [1]. In healthy individuals, plasma carnitine levels range from 30 to 89 μM [2]. Dietary intake of carnitine varies widely, generally ranging from < 11 to ~150 mg per day in a 70 kg individual [3]. Various aspects of carnitine supplementation have been studied [4–6]. In contrast, acetylcarnitine, a key short-chain ester of carnitine, has received significantly less attention. Acetylcarnitine serves a dual role by providing both carnitine and acetyl groups [7]. As a source of acetyl groups, acetylcarnitine may, under appropriate conditions, serve as a precursor for acetylcholine synthesis [8, 9]. Therefore, acetylcarnitine has been investigated as a potential treatment for various neurological disorders. A meta-analysis revealed that doses of 1.5 to 3 g/day administered over 3 to 12 months had positive effects in improving mild cognitive impairment or preventing its progression [10]. Additionally, supplementation with acetylcarnitine at a dose of 2 g/day was shown to increase circulating acetylcarnitine levels by 43% [3]. Studies examining blood acetylcarnitine levels in patients with chronic fatigue syndrome or during the post-COVID-19 period reported no significant differences compared with the levels in healthy individuals [11–13]. On the other hand, some studies have reported significantly elevated acetylcarnitine concentrations in patients who experienced heart failure [14, 15].

When consumed in excess, carnitine reaches and is metabolized by the gut microbiota [16]. In the gut, carnitine is metabolized to γ-butyrobetaine (GBB), which is subsequently converted to trimethylamine (TMA) by distinct bacterial species [17, 18]. TMA is effectively absorbed and then metabolized to trimethylamine N-oxide (TMAO) in the liver through the action of flavin-containing monooxygenases (FMOs) [19]. Increased levels of TMAO can be indicative of metabolic diseases such as atherosclerosis or diabetes, and the increase can be correlated with the severity of heart failure [17, 19–22] and mortality in older adults [23]. On the other hand, diets containing TMAO-rich fish products [24, 25] are associated with reduced risks of cardiovascular and metabolic diseases [26]. While it is unclear whether TMAO is directly involved in cardiometabolic diseases, its potential negative effects should be considered.

Although the use of carnitine as a dietary supplement has been extensively studied, there is limited information on the bioavailability of acetylcarnitine in healthy individuals, and its metabolism and impact on increased TMAO concentrations in humans remain unexplored. Our study provides novel insights into the bioavailability, metabolism, and excretion of supplements compared with those containing carnitine.

2 | Methods

2.1 | Participants

The participants were subjected to an interventional study approved by the Riga Eastern Clinical University Hospital Support Fund Medical and Biomedical Research Ethics Committee (opinion No. 18/2022, May 12, 2022). Before the study was conducted, the volunteers submitted information about their sex, age, and history of chronic and acute diseases. The exclusion criteria were as follows: diabetes mellitus, kidney or liver dysfunction, and acute inflammation. Volunteers completed a questionnaire in which they provided information about their eating habits for the previous 2 days to exclude the interactions of food with the research results. The main dietary exclusion criterion was the intake of products of fish origin, including fish oil, within the 2 days before the beginning of the study. None of the participants were vegan or vegetarian. No reported health-related or other defined criteria exclusions were recorded. All volunteers provided written informed consent to participate. All data were pseudonymized during collection and analyzed anonymously to protect participant identities.

2.2 | Study Design

A total of five open-label, randomized pharmacokinetic studies, including both single- and multiple-ascending-dose (SAD/MAD) designs, were conducted in healthy volunteers using various doses of carnitine and acetylcarnitine. An overview of the study design is presented in Table 1. At the onset of the study, 10 mL of fasting (at least 8 h of overnight fasting) blood samples, as well as morning urine samples, were collected and used to measure carnitine, acetylcarnitine, and other biochemical parameters at the beginning of the study. Shortly after initial blood sampling, the study participants received a supplement containing L-carnitine or acetylcarnitine. For the rest of the time points, 200 μL of blood were collected (Figure 1).

2.3 | Dosage Information

For the high-dose carnitine clinical experiment, six 500 mg capsules were used together, totaling 3000 mg of L-carnitine tartrate, of which 1560 mg was L-carnitine according to the manufacturer's description (L-carnitine capsules, 500 mg, Vplab, Latvia). After the experiment, the actual L-carnitine content in the capsule was measured via the same analytical method used for the carnitine measurements in the samples. The analysis of the capsules revealed that the actual carnitine content was 86% of 1560 mg, so the actual dose was equivalent to 1346 mg of L-carnitine. For the low-dose carnitine clinical experiment, the same capsules were used, but the participants took only two capsules, a total of 520 mg of carnitine according to the manufacturer's description (actual measured dose: 449 mg carnitine). For the repeated-dose clinical experiment, L-carnitine powder was analyzed before the experiment. To ensure that each dose contained 1500 mg of L-carnitine, the participants consumed 2.42 g of supplement powder (L-carnitine powder, OstroVit, Poland) once a day for 7 days. For the high-dose acetylcarnitine clinical experiment, three 500 mg capsules were taken together, for a total of 1500 mg of acetylcarnitine (acetyl L-carnitine, 500 mg, Ultra Vit, Latvia). Analysis of the capsules confirmed that the content of free-form acetylcarnitine corresponded to the labeled amount of 500 mg per capsule (approximately 104%). For the low-dose acetylcarnitine clinical experiment, the same supplement capsules were used, but only one capsule was taken. In every study, after fasting blood sample collection, supplements were consumed in the morning with a glass of water (200 mL).

2.4 | Determination of Carnitine, Acylcarnitine, TMAO, and GBB Levels in Plasma and Urine Samples

A liquid chromatography–tandem mass spectrometry (LC–MS/MS) method with a simple one-step protein precipitation using acetonitrile/methanol (3:1, v/v) was applied for the quantitative determination of carnitine, acetylcarnitine, TMAO, and GBB levels in plasma and urine samples, as previously described [27], with some modifications. Specifically, the method was extended by including acetylcarnitine quantification, and analysis was performed on a Shimadzu LCMS-8060NX system (Shimadzu, Kyoto, Japan). Plasma and urine samples and standards (20 μL) were mixed with 480 μL of internal standard solution, vortexed for 5 s, and centrifuged at 10,000 × *g* for 10 min at room temperature. The internal standard (IS) was 3-(2,2-dimethyl-2-prop-1-yl-hydrazinium)propionate, prepared in acetonitrile/methanol (3:1, v/v) at a concentration of 50 ng/mL. The supernatant was diluted 10-fold with acetonitrile/methanol (3:1, v/v) and transferred into LC vials for subsequent LC–MS/MS analysis. Chromatographic separation was performed on a Waters Acquity UPLC BEH HILIC column (2.1 mm × 50 mm, 1.7 μm). The mobile phase

TABLE 1 | Summary of performed clinical studies.

Study	Dose	Participants	Average age ± SEM (range)	Average BMI (range)	Supplement
Carnitine high	Single, 1500 mg	6 M/6 F	36 ± 3 (23–60)	24 (19–30)	L-carnitine capsules 500 mg, Vplab
Carnitine low	Single, 500 mg	5 M/6 F	36 ± 4 (23–60)	24 (19–30)	L-carnitine capsules 500 mg, Vplab
Acetylcarnitine high	Single, 1500 mg	5 M/7 F	38 ± 4 (24–60)	26 (19–29)	Acetyl L-carnitine 500 mg, Ultra Vit
Acetylcarnitine low	Single, 500 mg	1 M/5 F	30 ± 4 (23–44)	24 (19–29)	Acetyl L-carnitine 500 mg, Ultra Vit
Carnitine repeated	7 doses, 1500 mg	3 M/3 F	37 ± 4 (24–47)	23 (21–29)	L-carnitine powder, OstroVit

Note: Supplement doses are shown in milligrams (mg)
Abbreviations: F, female; M, male.

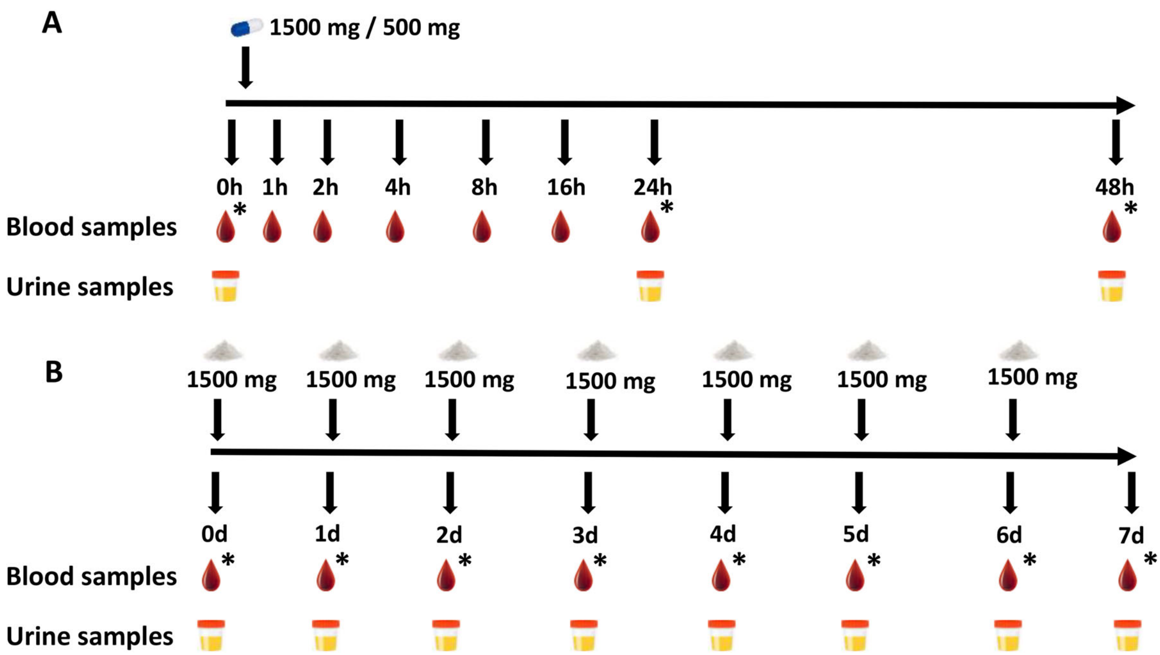

FIGURE 1 | Design scheme for single-dose (A) and repeated-dose (B) studies. *Time points where blood samples were taken in the fasted state.

consisted of acetonitrile (ACN) and 10 mM ammonium acetate in water (pH 4). Gradient elution was applied as follows: 0 min, 75% ACN; 2.5 min, 55% ACN; 4 min, 55% ACN; 4.5 min, 75% ACN; and 6 min, 75% ACN. The column temperature was maintained at 40°C, and the flow rate was set at 0.25 mL/min. The analysis was carried out in positive electrospray ionization (ESI) mode. The optimized mass spectrometric parameters were as follows: nebulizing gas flow at 3 L/min, heating gas flow at 15 L/min, interface temperature 400°C, desolvation line temperature 250°C, heat block temperature 400°C, desolvation temperature 650°C, and drying gas flow at 3 L/min. The optimized MRM transitions, along with Q1/Q3 voltages and collision energies (CE), were as follows: carnitine, m/z 162/103 (Q1: −29 V, CE: −18 V, Q3: −26 V); GBB, m/z 146/87 (Q1: −16 V, CE: −16 V, Q3: −18 V); TMAO, m/z 76/58 (Q1: −13 V, CE: −22 V, Q3: −21 V); acetylcarnitine, m/z 204/85 (Q1: −19 V, CE: −20 V, Q3: −18 V); and IS, m/z 175/58 (Q1: −20 V, CE: −21 V, Q3: −20 V). LabSolutions software (version 5.99; Shimadzu) was used for data acquisition and processing.

The quantitative determination of medium- and long-chain acylcarnitine levels was performed in plasma samples using UPLC–MS/MS as previously described [28]. Chromatographic separation was performed on a Waters Acquity UPLC BEH HILIC column (2.1 × 100 mm, 1.7 µm) using a Waters Acquity UPLC H-Class system coupled to a Waters Xevo TQ-S tandem mass spectrometer (Waters, Milford, MA, USA). The mobile phase consisted of acetonitrile and 10 mM ammonium acetate with 0.2% formic acid in the water, and was applied using a gradient elution as follows: 0 min–90% ACN, 6 min–80% ACN, 7.3 min–35% ACN, 8 min–35% ACN, 8.3 min–90% ACN, and 10 min–90% ACN. The column temperature was maintained at 30°C, and the flow rate was set at 0.5 mL/min. The analysis was carried out in positive electrospray ionization (ESI+) mode using multiple reaction monitoring (MRM). Data acquisition and processing were performed using Waters MassLynx 4.1 and TargetLynx software.

2.5 | Data Analysis

Data distribution was assessed for normality using the Shapiro–Wilk test. As the data were not normally distributed, results from volunteer samples are expressed as medians with interquartile ranges (IQR), and nonparametric methods were applied. For comparing repeated measures with baseline, the Friedman test was used. In cases where the Friedman test indicated statistically significant differences, Dunn's multiple comparisons test was performed post hoc to identify specific group differences. Delta area under the curve (ΔAUC) values were analyzed using the Kruskal–Wallis test, followed by Dunn's multiple comparisons test for pairwise comparisons between groups. All analyses were conducted using GraphPad Prism 10.3.0, and p-values less than 0.05 were considered statistically significant. Microsoft 365 Excel was used for data collection.

3 | Results

3.1 | Carnitine, Acetylcarnitine, and TMAO Plasma Concentrations After Single-Dose Administration

The plasma concentrations of acetylcarnitine, carnitine, and the main metabolite TMAO, as well as the ΔAUC values, are presented in Figure 2 (Measured compound baseline values are in supplementary information). After acetylcarnitine supplementation, plasma concentrations peaked at 4 h. With the 0.5 g dose, median levels rose from 4.7 µM (IQR: 4.3–5.7) at baseline to 5.5 µM (IQR: 4.7–7.1). With the 1.5 g dose, concentrations increased from a median of 9.4 µM (IQR: 7.9–11.7) to 12.5 µM (IQR: 9.8–14.4). However, the increase was not statistically significant (Figure 2A). Following administration of 1.5 g of carnitine, plasma acetylcarnitine concentrations nearly doubled from 12.1 µM (IQR: 8.8–14.7) at baseline to 20.9 µM (IQR: 17.1–25.9) in 4 h. A smaller rise was observed with a 0.5 g carnitine dose, from 5.6 µM (IQR: 5.0–6.4) to 8.6 µM (IQR: 5.7–12.6) over the same period. Both doses of carnitine induced greater increases in acetylcarnitine levels than did the corresponding doses of acetylcarnitine.

The plasma acetylcarnitine ΔAUC values were not significantly different between the tested carnitine and acetylcarnitine supplementation groups. The ΔAUC for acetylcarnitine following a 1.5 g dose of carnitine was nearly identical to that observed after 1.5 g of acetylcarnitine. A 0.5 g dose of carnitine resulted in a ΔAUC of 49.1 ± 14.2 µM h, whereas the lower acetylcarnitine dose led to a markedly lower ΔAUC of 19.4 ± 8.8 µM h (Figure 2B).

After a 1.5 g dose of carnitine was given, the plasma carnitine concentration gradually increased, peaking at a median of 60.9 µM (IQR: 58–65) at the 8-h mark from a median baseline of 44.4 µM (IQR: 34.7–47.2). Carnitine levels then gradually declined but remained significantly elevated for at least 24 h (Figure 2C). A lower dose of carnitine (0.5 g) increased plasma carnitine concentrations from a median of 36.3 µM (IQR: 25.4–41.1) to 43.4 µM (IQR: 35.4–46.6) at 4 h. Similarly, the higher acetylcarnitine dose (1.5 g) increased plasma carnitine from 35.3 µM (IQR: 25.7–48.0) at baseline to 43.9 µM (IQR: 35.8–47.9) at 16 h, an effect comparable to that of 0.5 g of carnitine.

The highest plasma carnitine ΔAUC over 48 h was observed for the 1.5 g carnitine dose, which exhibited significantly higher values than those observed with acetylcarnitine supplementation (Figure 2D). The ΔAUC after administration of 0.5 g of carnitine was two times lower than that after administration of 1.5 g of carnitine. Despite the 3-fold difference in dose, the lower carnitine dose resulted in a relatively high carnitine increase in plasma. When comparing the 1.5 g doses of carnitine and acetylcarnitine, the acetylcarnitine ΔAUC showed a 3-fold decrease. The 0.5 g acetylcarnitine dose resulted in a minimal plasma carnitine ΔAUC of only 12.3 ± 7.2 µM h.

The baseline plasma TMAO among the study participants ranged from 0.94 to 7.76 µM. Both carnitine supplementation and acetylcarnitine supplementation led to a substantial increase in plasma TMAO levels. A 1.5 g dose of carnitine increased the TMAO concentration from a median of 2.2 µM (IQR: 1.9–3.2) to a median of 54.9 µM (IQR: 30.1–65.6), with peak levels occurring 24 h after intake (Figure 2E). A lower carnitine dose (0.5 g) increased the TMAO concentration from a median of 3.2 µM (IQR: 1.3–4.3) to a median of 26.3 µM (IQR: 20.7–29.8), reaching 37% of the peak concentration observed with the 1.5 g dose, which is consistent with the dose difference.

Acetylcarnitine supplementation also resulted in TMAO production, with peak concentrations observed 16 h post-intake. A 1.5 g dose of acetylcarnitine significantly increased the TMAO concentration from a median of 2.2 µM (IQR: 1.9–3.1) to a median of 47 µM (IQR: 22.4–57.3), whereas a 0.5 g dose increased the TMAO concentration from a median of 3.2 µM (IQR: 1.3–4.3) to a median of 26.3 µM (IQR: 20.7–29.8). The molar fraction of carnitine in the acetylcarnitine supplements, after deacetylation, accounts for approximately 79% of the total supplement. Considering this and adjusting for carnitine content, the ΔAUC values of TMAO in plasma for 1.5 g of acetylcarnitine (978 µM h, adjusted to 1183 µM h) were closely aligned with those observed for the 1.5 g carnitine supplements (1191 µM h).

The highest plasma TMAO ΔAUC values were observed following the administration of 1.5 g of carnitine and acetylcarnitine (Figure 2F). A comparison of the high- and low-dose groups revealed significantly lower TMAO ΔAUC values in the low-dose supplementation groups.

3.2 | Carnitine, Acetylcarnitine, and TMAO Concentrations in Urine After Single-Dose Administration

The urinary concentrations at different time points and the corresponding ΔAUC values are presented in Figure 3. The most pronounced increase in acetylcarnitine excretion was observed following the intake of 1.5 g of carnitine, with the highest measured urinary acetylcarnitine concentrations at 24 h. In contrast, other supplements induced only minor increases in acetylcarnitine excretion, which fell within the range of

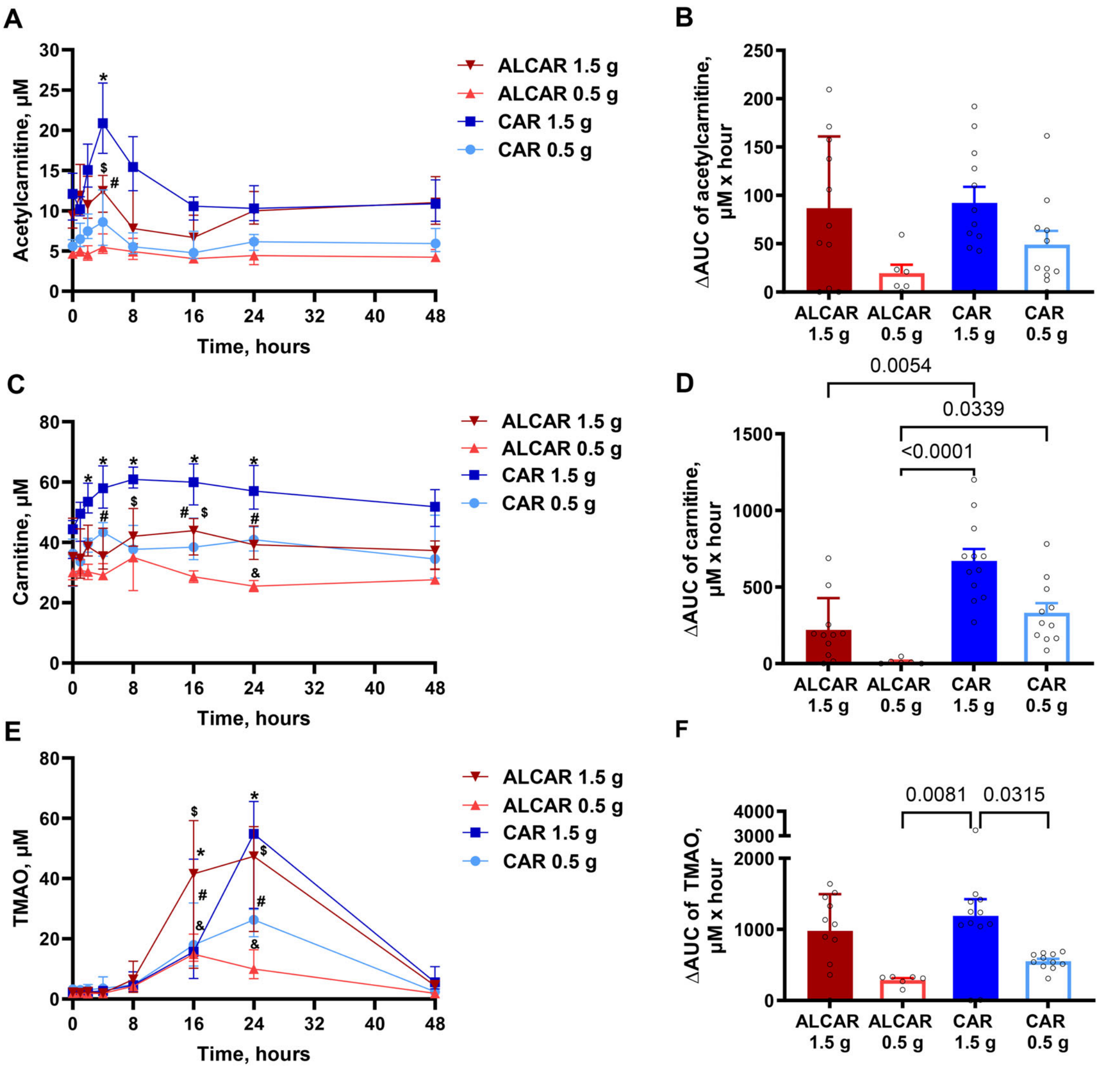

FIGURE 2 | Changes in plasma acetylcarnitine (A, B), carnitine (C, D), and TMAO (E, F) concentrations and the ΔAUC after carnitine (CAR)- and acetylcarnitine (ALCAR)-containing supplement intake at doses of 1.5 and 0.5 g. Blood samples were taken before supplement intake and 2, 4, 8, 16, 24, and 48 h after supplement intake. ΔAUC values were calculated from the increase in measured concentrations above baseline (B, D, F). Data are expressed as medians from measurements in the following number of volunteers: 12 (CAR 1.5), 11 (CAR 0.5), 11 (ALCAR 1.5), 6 (ALCAR 0.5), and IQR. *—carnitine, 1.5 g dose, $p < 0.05$; #—carnitine, 0.5 g dose, $p < 0.05$; $—acetylcarnitine, 1.5 g dose, $p < 0.05$; &—acetylcarnitine, 0.5 g dose, $p < 0.05$. ALCAR, acetylcarnitine supplements; CAR, carnitine supplements; IQR, interquartile range.

natural fluctuations in urinary acetylcarnitine concentrations (Figure 3A). The ΔAUC for urinary acetylcarnitine concentrations was significantly greater in the 1.5 g carnitine group than in the 1.5 g acetylcarnitine group. The ΔAUC values for the 0.5 g acetylcarnitine and 0.5 g carnitine groups were comparable (Figure 3B). Urinary analysis confirmed that a substantial portion of the administered carnitine was excreted via urine. The most pronounced increase in the urinary carnitine concentration was observed following the intake of 1.5 g of carnitine, where the concentration rose from a baseline median of 40 µM (IQR: 20–99) to 292.2 µM (IQR: 92–629) at 24 h, and remaining elevated at 48 h (Figure 3C). The 0.5 g carnitine dose also increased urinary carnitine concentrations at both measurement time points, with the highest concentration at 48 h, although this represented a relatively modest increase over baseline (baseline: median 20 µM, IQR: 10–88; 48-h: median 60 µM, IQR: 19–130).

Acetylcarnitine supplementation with both doses similarly elevated urinary carnitine concentrations. The 1.5 g acetylcarnitine dose increased urinary carnitine levels from a median of 33 µM (IQR: 13–195) to 91 µM (IQR: 18–406) at 24 h. The 0.5 g acetylcarnitine dose had a similar effect on urinary carnitine; it increased from a baseline median of 56 µM (IQR: 27–101) to 123 µM (IQR: 42–135) after 24 h.

The ΔAUC for urinary carnitine concentrations differed significantly between the 1.5 g carnitine dose group and all the other supplementation groups. The ΔAUC following 1.5 g carnitine intake was 8659 ± 2126 µM h, significantly exceeding the ΔAUC observed for the 1.5 g acetylcarnitine group (Figure 3D). These findings indicate that higher plasma carnitine levels also result in significantly greater urinary elimination. The ΔAUC values for the 0.5 g carnitine and 0.5 g acetylcarnitine groups were similar.

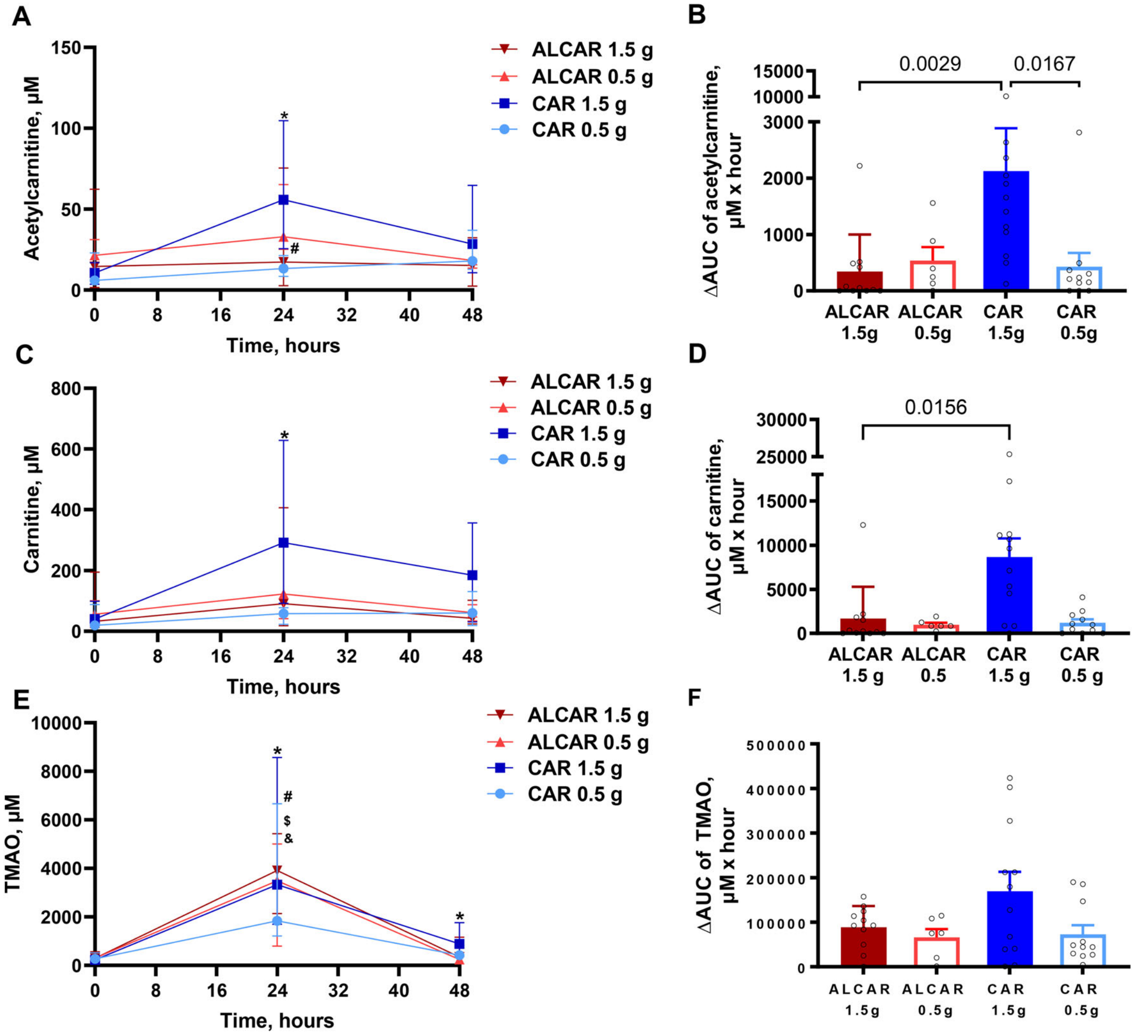

FIGURE 3 | Changes in urinary concentrations of acetylcarnitine (A, B), carnitine (C, D), and TMAO (E, F) and the ΔAUC after carnitine (CAR)- and acetylcarnitine (ALCAR)-containing supplement intake at doses of 1.5 and 0.5 g. Urine samples were obtained before supplement intake and 24 and 48 h after supplement intake. ΔAUC values were calculated from the increase in measured concentrations above baseline (B, D, F). Data are expressed as medians from measurements in the following number of volunteers: 12 (CAR 1.5), 11 (CAR 0.5), 11 (ALCAR 1.5), 6 (ALCAR 0.5), and IQR. *—Carnitine, 1.5 g dose, $p < 0.05$; #—carnitine, 0.5 g dose, $p < 0.05$; \$—acetylcarnitine, 1.5 g dose, $p < 0.05$; &—acetylcarnitine, 0.5 g dose, $p < 0.05$. ALCAR, acetylcarnitine supplements; CAR, carnitine supplements; IQR, interquartile range.

Urinary TMAO measurements revealed a significant increase in TMAO production and subsequent excretion following both carnitine and acetylcarnitine supplementation (Figure 3E). Intake of 1.5 g of carnitine caused a sharp increase in urinary TMAO concentrations, rising from a baseline median of 229 μM (IQR: 118–265) to 3329 μM (IQR: 1692–8568) after 24 h. TMAO levels remained elevated after 48 h (median: 886, IQR: 378–1764). A lower dose of carnitine (0.5 g) also induced a substantial TMAO increase from a baseline median of 272 μM (IQR: 164–519) to a peak of 1835 μM (IQR: 1213–6659) after 24 h.

A similar increase in urinary TMAO concentrations was observed following acetylcarnitine supplementation. The 1.5 g dose increased TMAO from a median of 294 μM (IQR: 160–560) to a median of 3912 μM (IQR: 2137–5432) at 24 h, whereas the 0.5 g dose increased TMAO from a median of 306 μM (IQR: 210–369) to a median of 3484 μM (IQR: 797–5007). By 48 h, TMAO concentrations had returned close to baseline for both acetylcarnitine doses.

The highest urinary TMAO ΔAUC was observed in the 1.5 g carnitine group, although this value was not significantly different from those for the other supplementation groups. Comparable urinary TMAO ΔAUC values were recorded across the remaining supplementation groups (Figure 3F).

3.3 | Carnitine and TMAO Concentration Measurements After Repeated Intake of the Carnitine Supplement

Following repeated intake of a 1.5 g dose of carnitine for 7 days, no significant increase in the plasma carnitine concentration was detected on any day of the experiment compared with that at baseline (Figure 4). The highest plasma carnitine concentration (median: 49 μM, IQR: 41–59) was recorded on Day 5, reflecting only a slight increase from the baseline concentration (median: 37 μM, IQR: 33–53). By the end of the 7-day treatment, the plasma carnitine concentration had reached a median of 45 μM (IQR:

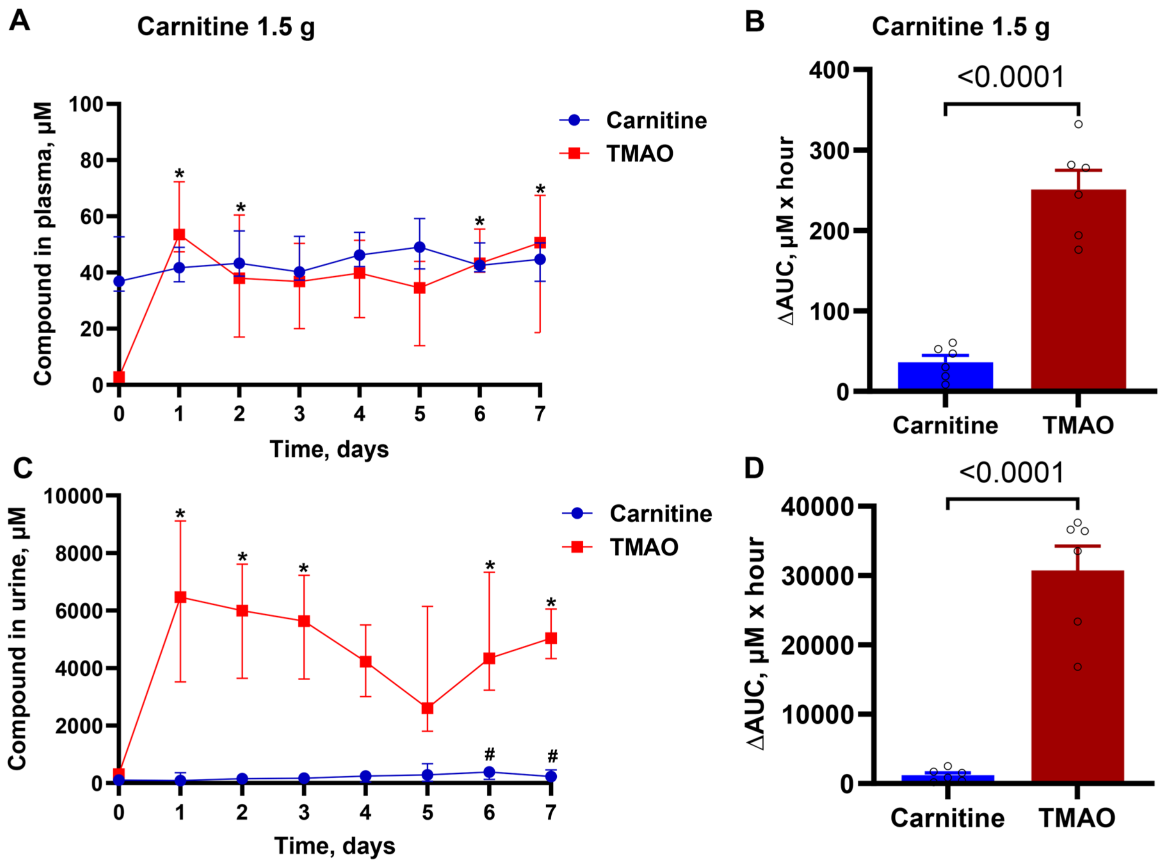

FIGURE 4 | Carnitine and TMAO concentrations and corresponding ΔAUC values in plasma (A, B) and urine (C, D) during the 7 days of continued carnitine administration. Plasma and urine samples were collected 24 h after the intake of carnitine. ΔAUC values were calculated from the increase in measured concentrations above baseline (B, D). Data are expressed as medians from measurements in the 6 volunteers and IQR. *p* values to compare the compound ΔAUC values were calculated using the unpaired *t*-test. IQR, interquartile range. *—TMAO, $p < 0.05$.

36.8–50.6), corresponding to an increase of 22% over the baseline value.

In contrast, the plasma TMAO concentration increased markedly after the first dose of carnitine. A significant 18-fold increase was observed, with the TMAO concentration reaching a median of 53.6 µM (IQR: 47.4–72.3) compared with the baseline concentration median of 2.8 µM (IQR: 2.5–4.6). The TMAO concentration remained elevated throughout the treatment period, with a slight decrease observed on Day 5. The ΔAUC of TMAO was 7 times greater than that of plasma carnitine.

Carnitine excretion increased progressively over the treatment period. At baseline, the urinary carnitine concentration was 103 µM (IQR: 12–125), peaking at 382 µM (IQR: 125–533) on Day 6.

Like in the single-dose experiment, repeated carnitine administration led to a rapid and sustained increase in urinary TMAO excretion. Following the first dose, the urinary TMAO concentration increased from median 306 µM (IQR: 128–472) at baseline to a median 6464 µM (IQR: 3519–9116). The TMAO level remained significantly elevated throughout the treatment period. The ΔAUC of urinary TMAO was 18 times greater than that of urinary carnitine.

3.4 | Measurements of γ-butyrobetaine (GBB) and Acylcarnitine Levels After Single-Dose Intake of Carnitine and Acetylcarnitine Supplements

The consumption of carnitine led to increased GBB concentrations in both plasma and urine. The highest increase in plasma GBB levels was observed following the administration of a 1.5 g dose of carnitine. Compared with the TMAO concentration, the GBB concentration peaked earlier, at 16 h post-carnitine intake. A lower carnitine dose (0.5 g) increased the plasma GBB level to a median of 1.24 µM (IQR: 1.10–1.40). In contrast, acetylcarnitine intake resulted in a significantly smaller increase in GBB level than either carnitine dose did (Figure 5A). Following acetylcarnitine intake, the peak plasma GBB concentration was also observed at 16 h, with both doses inducing a modest increase of 10%.

The highest ΔAUC for plasma GBB over 48 h was recorded for the 1.5 g carnitine dose, and this increase was significantly greater than that for the other carnitine and acetylcarnitine supplements (Figure 5B). The 0.5 g carnitine dose resulted in a ΔAUC 48% lower than that for the 1.5 g carnitine dose. When the 1.5 g carnitine and 1.5 g acetylcarnitine groups were compared, the latter presented a ΔAUC that was three times lower. The low-dose acetylcarnitine supplement yielded only a minimal increase.

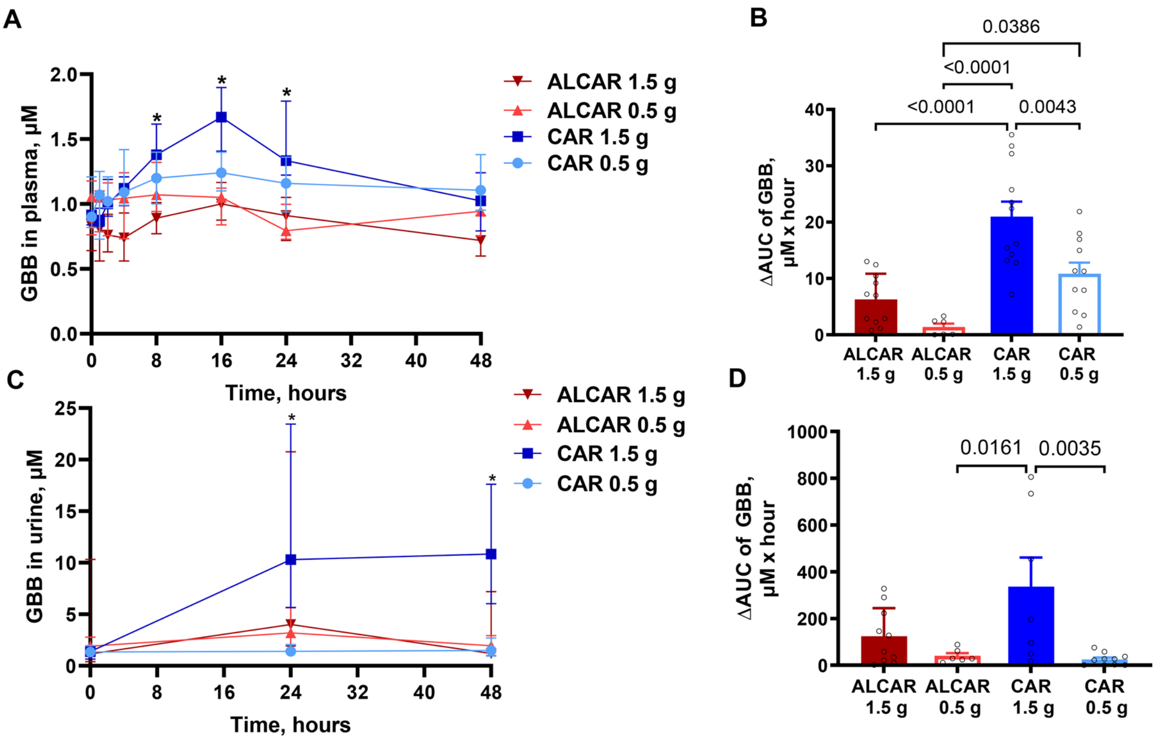

FIGURE 5 | Changes in GBB concentrations in plasma (A, B) and urine (C, D), as well as the ΔAUC, after the intake of supplements containing carnitine (CAR) and acetylcarnitine (ALCAR) at doses of 1.5 and 0.5 g. Blood samples were taken before supplement intake and 2, 4, 8, 16, 24, and 48 h after supplement intake. Urine samples were obtained before supplement intake and 24 and 48 h after supplement intake. ΔAUC values were calculated from the increase in measured concentrations above baseline (B, D). Data are expressed as median from measurements in the following number of volunteers: 12 (CAR 1.5), 11 (CAR 0.5), 11 (ALCAR 1.5), 6 (ALCAR 0.5), and IQR. *—carnitine, 1.5 g dose, $p < 0.05$. ALCAR, acetylcarnitine supplements; CAR, carnitine supplements; IQR, interquartile range.

Similar trends were observed in the urine samples. The 1.5 g carnitine dose increased the urinary GBB concentration from a median of 1.4 μM (IQR: 0.66–1.73) to a median of 10.3 μM (IQR: 5.7–23.5) at 24 h postintake, with the level remaining elevated even at 48 h (Figure 5C). Acetylcarnitine supplementation at the same dose also increased the urinary GBB concentration, which reached a median of 4.0 μM (IQR: 1.9–20.8) at 24 h. Lower doses of both supplements had only a minor effect on urinary GBB concentrations.

The ΔAUC values for urinary GBB were significantly greater following 1.5 g carnitine supplementation than following 1.5 g acetylcarnitine supplementation (Figure 5D). However, in contrast to findings in plasma, the 0.5 g acetylcarnitine dose resulted in a higher excreted ΔAUC than the 0.5 g carnitine supplement did.

Figure 6 shows the plasma concentrations of long-chain (C14–C18) and medium-chain (C5–C12) acylcarnitines following high-dose supplementation with acetylcarnitine and carnitine. Overall, neither supplement had a significant effect on long- or medium-chain acylcarnitine levels.

Two hours after high-dose acetylcarnitine intake, the medium-chain acylcarnitine concentration increased by 10%. This initial increase was followed by a slight decrease below baseline, but the concentration subsequently peaked after 24 h. In contrast, long-chain acylcarnitine levels remained relatively stable across time points, although a 16% decrease below baseline was observed at 8 h (Figure 6A). Subsequently, concentrations gradually increased, peaking at 48 h post-supplementation.

In contrast to acetylcarnitine supplementation, carnitine supplementation led to an initial 25% reduction in medium-chain acylcarnitine concentration within 2 h. The concentration then increased at 4 h but subsequently decreased by 16 h (Figure 6B). After 24 h, the concentration returned nearly to baseline.

Following high-dose carnitine supplementation, the long-chain acylcarnitine concentration showed an overall 11% reduction from the baseline average (50 μM) throughout the study, with the most pronounced decrease (22%) occurring at 24 h (0.38 μM).

4 | Discussion

In this study, we demonstrate that the intake of acetylcarnitine or carnitine supplements at the manufacturer-recommended doses results in a lower bioavailability than previously anticipated [3, 29]. Manufacturers typically recommend a daily intake of 1.5 g of carnitine or acetylcarnitine, significantly exceeding the 11–150 mg of carnitine obtainable from dietary sources. Previous studies reported that a 2 g oral dose of carnitine yielded a bioavailability of 9%–25% in healthy subjects on a low-carnitine diet [29]. In our tests involving healthy volunteers on a normal diet and doses of 0.5–1.5 g of acetylcarnitine and carnitine supplements, we

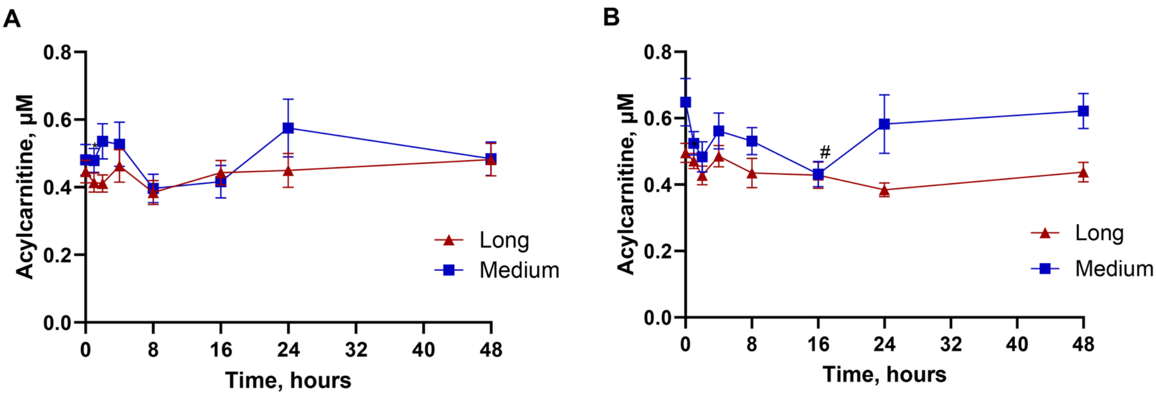

FIGURE 6 | Changes in long- and medium-chain acylcarnitine concentrations in plasma after (A) 1.5 g acetylcarnitine supplement intake and (B) 1.5 g carnitine supplement. Blood samples were taken before supplement intake and 2, 4, 8, 16, 24, and 48 h after supplement intake. Data are expressed as the average from measurements in the following number of volunteers: 12 (CAR 1.5), 11 (ALCAR 1.5) ± SEMs. #—medium chain acylcarnitine, $p < 0.05$ compared with baseline. SEM, standard error of the mean.

found that the bioavailability was less than 5%. Moreover, the bioavailability of acetylcarnitine was substantially lower than that of carnitine.

Carnitine and acetylcarnitine are lost from circulation through urine and must be replenished through dietary intake and biosynthesis. In an omnivorous adult, biosynthesis contributes approximately 11.3–22.6 mg of carnitine per day for a 70 kg individual [3, 30]. In this study, we found that volunteers excreted approximately 10 mg of acetylcarnitine and 30 mg of carnitine daily. A typical omnivorous diet provides enough carnitine to compensate for its urinary losses [31], making carnitine biosynthesis a minor contributor to the overall carnitine supply. Instead, organic cation transport protein 2 (OCTN2) plays a key role in regulating carnitine and acetylcarnitine levels in body fluids and tissues [32]. OCTN2 mediates absorption in the small intestine, transport into tissues, and renal reabsorption. Accordingly, the intestinal absorption rates of carnitine and acetylcarnitine are similar to their renal reabsorption rates [33] and are limited by OCTN2 transporter capacity. As a result, dietary carnitine is efficiently absorbed at low concentrations in the intestine [3]. However, the relative bioavailability of high-dose carnitine supplements, which can reach millimolar concentrations in gastrointestinal fluids, is very low [34]. In this study, we demonstrated that not only carnitine but also acetylcarnitine supplements at doses of only 0.5–1.5 g exceeded the carnitine transporter capacity, resulting in highly limited uptake in the intestine. We also observed greater bioavailability at the 0.5 g dose than at the 1.5 g dose of the carnitine supplement. Thus, carnitine and acetylcarnitine uptake in the intestine was limited to less than 10% of the intake dose, which resulted in maximum increases in carnitine and acetylcarnitine concentrations in plasma of 47% and 43%, respectively, whereas the increase in plasma acetylcarnitine exposure (ΔAUC) was more than 7-fold lower than the corresponding increase in carnitine exposure at the same dose (1.5 g). The acetylcarnitine K_m for OCTN2 is 2 times lower than that for carnitine (18.5 vs. 4.3 µM) [35, 36], which explains the even lower acetylcarnitine bioavailability.

The carnitine concentration in plasma (30–60 µM) is regulated by the renal tubular reabsorption capacity. When the carnitine concentrations exceed this range, the excess carnitine is eliminated via the urine. After the intake of 1.5 g of carnitine, renal clearance increased to 1.33 L/h, which is 5.4 times greater than the baseline carnitine clearance of 0.278 L/h reported previously [37]. Increased clearance resulted in a 5.5-fold higher urine concentration and a 5-fold greater amount of eliminated carnitine than at baseline. We observed a different effect for acetylcarnitine; we detected a significant increase in the carnitine but not the acetylcarnitine concentration in urine. Therefore, a significant portion of the carnitine present in the bloodstream is excreted in the urine rather than being absorbed by tissues [37]. In previous studies, oral bioavailability was estimated by comparing plasma ΔAUC of intravenous and oral intake of carnitine supplements; however, considering intake-stimulated elimination of carnitine and acetylcarnitine in urine, the true net benefit of supplementation (uptake in tissues) is much lower than it was estimated before. We found that 5%–10% of carnitine is lost via urine, which is consistent with the previously reported 6% of the administered dose being eliminated via urine [38]. In the repeated administration study, we observed a gradually increasing elimination of carnitine via urine over a 7-day treatment period, suggesting a decrease in OCTN2 transport capacity in the presence of elevated carnitine levels. Overall, low OCTN2 transporter capacity might be a reason for low carnitine and acetylcarnitine intestinal uptake and renal tubular reabsorption.

In this study, we observed a rapid increase in acetylcarnitine concentrations in plasma after the intake of carnitine alone. This resulted in a greater increase in the acetylcarnitine concentration after the intake of carnitine than after the addition of the acetylcarnitine supplement. Since the acetylcarnitine concentration was also markedly elevated in urine (8-fold) after carnitine intake, we concluded that carnitine intake stimulates the elimination of acetyl groups (in the form of acetylcarnitine) from tissues to the urine. Additionally, after the intake of acetylcarnitine, we observed an increase in the urinary carnitine level, but this effect was less pronounced since the increase in the acetylcarnitine concentration in the plasma and urine after acetylcarnitine intake was relatively low. In the fasting state, acetylcarnitine efflux from the hepatosplanchnic bed has been previously measured [39]. Carnitine and acetylcarnitine compete for the same transporter; therefore, as acetylcarnitine appears outside tissue, higher carnitine concentrations in plasma and primary urine compete with acetylcarnitine for return into tissues and reabsorption in the kidneys. We also measured the levels of other acylcarnitine species to evaluate their possible interactions with carnitine and acetylcarnitine. The concentrations of medium-chain acylcarnitines decreased shortly after carnitine intake and then returned to baseline concentrations 24 h after carnitine intake. Overall, changes in medium- and long-chain acylcarnitine concentrations are more strongly affected by the fasting and postprandial status of volunteers than by the intake of carnitine.

Studies of the bioavailability and turnover of carnitine and acetylcarnitine are important for predicting the possible health effects of these supplements. In tissues with high fatty acid metabolism rates, such as the heart, muscle, and liver, carnitine concentrations are in the hundred micromolar to millimolar range [40]. These levels far exceed the K_m values for carnitine-requiring enzymes and transporters such as CPT1, CACT, CROT, and CRAT. At a substrate concentration that is much greater than the K_m [37] for the reaction, the enzyme operates at its maximum rate, and additional substrate supply has no apparent benefit. Overall, because of the high tissue levels and low oral bioavailability of carnitine, the intake of carnitine supplements cannot substantially change the carnitine levels in tissues; therefore, any effect of supplementation is unlikely.

Measurements of the concentrations of carnitine, acetylcarnitine, and microbiota-derived metabolites can provide an estimate of the time- and concentration-dependent microbial metabolism of these substances. The carnitine concentration in plasma peaked 8 h after intake of the supplement, after which the concentration of carnitine gradually decreased. The peak acetylcarnitine concentration in the plasma was observed 4 h after intake of the supplement, after which a rapid decrease to the baseline concentration was observed. These results indicate that carnitine and acetylcarnitine are absorbed mainly in the upper part of the gastrointestinal system. If supplements are not absorbed early in the gastrointestinal tract, carnitine and acetylcarnitine remain in the intestine much longer, and reach the anaerobic microbiota, which degrades carnitine and acetylcarnitine. Acetylcarnitine is converted to carnitine, which is subsequently converted to GBB and TMA by different microbiota species [41, 42]. In this study, we observed a small increase in plasma carnitine concentration, which peaked at 8 h after acetylcarnitine intake. Some of the carnitine produced by the microbiota from acetylcarnitine may be absorbed, whereas most is further degraded. GBB is produced from L-carnitine by different species, such as *Enterobacteriaceae, Escherichia coli (E. coli)*, and *Proteus mirabilis* [41, 42]. After carnitine intake, GBB appears in the plasma at 1 h and reaches a peak concentration 16 h after the intake of carnitine. Therefore, the degradation of carnitine to GBB starts relatively early. Like carnitine, GBB is degraded to TMA by *Enterobacteriaceae*, such as *E. coli, Salmonella typhimurium*, and *Proteus vulgaris* [43]. The TMAO levels in plasma and urine indicate that TMA is the main product of microbiota-driven degradation of carnitine and acetylcarnitine. We observed a relatively late appearance of TMAO, which confirms that carnitine is first degraded to TMA by anaerobic microbes, which are located in the lower part of the gastrointestinal system. The late appearance of TMAO can be attributed to the need for an additional metabolic step in the liver for the conversion of TMA to TMAO. Overall, excess carnitine and acetylcarnitine in food supplements are degraded to GBB and TMA and further metabolized to TMAO.

The TMAO measurements indicated that carnitine bioavailability is independent of the microbiota, as the main intestinal absorption occurs early while carnitine degradation by the microbiota occurs substantially later after the intake of supplements. TMAO was excreted in the urine within 48 h following the intake of acetylcarnitine and a lower (0.5 g) dose of carnitine. However, after administration of 1.5 g of carnitine, TMAO levels remained elevated and did not return to baseline levels within the same timeframe. We estimated the total TMAO excretion in urine and reported that up to 90% of the intake dose of carnitine and acetylcarnitine was eliminated as TMAO. In addition, 5%–10% of the supplement dose was eliminated via urine as carnitine and acetylcarnitine, which resulted in a remaining bioavailability of less than 5% of the dose.

The intake of carnitine and acylcarnitine supplements at a daily dose of 1.5 g increased the plasma levels of TMAO more than 30-fold, with concentrations reaching over 40–50 µM, whereas the average TMAO concentration at baseline was 2–3 µM.

The discussion regarding TMAO as a cardiometabolic risk marker is still ongoing. After the first evidence of the proatherosclerotic effects of TMAO and its prognostic value in patients with heart failure [19, 44], a positive relationship between the TMAO concentration and the development of various cardiovascular, metabolic, and neurological diseases, as well as cancer, was reported [45]. The clinically relevant detrimental plasma concentration of TMAO is not officially defined and varies across studies. A study that included 2595 participants and predicted increased risks for prevalent cardiovascular disease and incident major adverse cardiac events in subjects with high TMAO levels used median plasma levels of carnitine (46.8 µM) and TMAO (4.6 µM) to stratify subjects into “low” (< median) or “high” (≥ median) plasma concentration groups [16]. Our study revealed that such levels of TMAO can be reached after a single intake of 1.5 g of carnitine and acetylcarnitine. Moreover, after repeated intake of carnitine, both the carnitine and TMAO plasma concentrations plateaued at approximately 40 µM, which might pose health risks. On the other hand, some preclinical studies have reported protective effects of TMAO, such as cardioprotective effects during heart failure or protective effects by increasing diuresis and natriuresis [46–48]. Since both carnitine and acetylcarnitine intake increase TMAO levels, both supplements can be used to study the effects of TMAO under different health conditions.

4.1 | Limitations of the Study

A key limitation of our study is that we were unable to collect complete 24-h urine samples from all participants, which may have led to an underestimation of total compound excretion. As a result, the calculated urinary excretion and AUC values should be interpreted as approximate indicators rather than precise totals. Additionally, our study did not assess individual differences in gut microbiota composition, which influences TMAO production and may account for individual variability in response to carnitine and acetylcarnitine supplementation.

Finally, the relatively short duration of observation (48 h) limits our ability to draw conclusions about the long-term metabolic fate and health implications of repeated acetylcarnitine supplement intake.

5 | Conclusion

Our study provides new insights into acetylcarnitine-containing supplements, highlighting their low bioavailability, high excretion rates, and extensive metabolism into TMAO. It also confirms the limited effectiveness of carnitine supplementation, because carnitine is also mostly converted into TMAO, which reached very high levels in plasma and urine after supplementation. These findings contribute to a deeper understanding of the turnover dynamics of carnitine and acetylcarnitine, including their absorption, distribution, microbial transformation, and elimination.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available on request due to privacy/ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: mnfr70316-sup-0001-SuppMat.docx.